THE NORMALIZATION OF A BASE IN A NORMAL SOURCE IS A DISJOINT UNION OF INTEGRAL NORMAL SCHEMES WITH DOMINANT STRUCTURE MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-normal-normalization is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 14169-14180 and proof span 14182-14218 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof twice refers to a topological argument that the author leaves out ('which we omit' and 'omitted'), once for the local finiteness of irreducible components of the normalization and once for the bijection on irreducible components; both phrases are kept verbatim, and in each case the geometric input (the description of the normalization over affine opens) is established in the counted span.

Editorial note. 7 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The compatibility of normalization with localization, cited from the same chapter. Cited at proof lines 14183-14185. Reduces the statement over an affine open of the base to the normalization of that ring in the preimage.
- A normal scheme whose quasi-compact opens have finitely many irreducible components is a disjoint union of normal integral schemes, cited from the properties chapter. Cited at proof lines 14189-14192 and 14208-14211. Decomposes the preimage of an affine open into finitely many normal integral pieces and gives the global conclusion.
- The compatibility of normalization with disjoint unions, established by a previous card of this lane, and the description of the normalization by integral closures. Cited at proof lines 14193-14196 and 14200-14203. Splits the normalization over the affine open into the normalizations in the individual pieces.

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- The ring of global functions of a normal integral affine scheme is a normal domain, and the integral closure of a ring in a normal domain is a normal domain. Cited at proof lines 14197-14206. Shows that each piece of the normalization is again a normal integral scheme.

Source reference. Original lemma 034L: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4096–4096. Same-chapter reference used in the retained author text at source line(s) 14200. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 035F: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4126–4126. Same-chapter reference used in the retained author text at source line(s) 13884, 13969, 14199, 14489. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 03GO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4535–4535. Same-chapter reference used in the retained author text at source line(s) 14192, 14455. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 035K: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4131–4131. Same-chapter reference used in the retained author text at source line(s) 14087, 14185, 14659. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 033J: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4058–4058. Same-chapter reference used in the retained author text at source line(s) 14207. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0358: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4119–4119. Same-chapter reference used in the retained author text at source line(s) 14195. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0357: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4118–4118. Same-chapter reference used in the retained author text at source line(s) 14191, 14212, 14660. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $f : Y \to X$ be a quasi-compact and quasi-separated morphism of schemes. Let X' be the normalization of X in Y. Assume*

(1) *Y is a normal scheme,*

(2) *quasi-compact opens of Y have finitely many irreducible components.*

Then X' is a disjoint union of integral normal schemes. Moreover, the morphism $Y \to X'$ is dominant and induces a bijection of irreducible components.

Proof. Let $U \subset X$ be an affine open. Consider the inverse image U' of U in X'. Set $V = f^{-1}(U)$. By Lemma 035K we $V \to U' \to U$ is the normalization of U in V. Say $U = \mathrm{Spec}(A)$. Then V is quasi-compact, and hence has a finite number of irreducible components by assumption. Hence $V = \coprod_{i=1,\ldots,n} V_i$ is a finite disjoint union of normal integral schemes by Properties, Lemma 0357. By Lemma 03GO we see that $U' = \coprod_{i=1,\ldots,n} U'_i$, where U'_i is the normalization of U in V_i. By Properties, Lemma 0358 we see that $B_i = \Gamma(V_i, \mathcal{O}_{V_i})$ is a normal domain. Note that $U'_i = \mathrm{Spec}(A'_i)$, where $A'_i \subset B_i$ is the integral closure of A in B_i, see Lemma 035F. By Algebra, Lemma 034L we see that $A'_i \subset B_i$ is a normal domain. Hence $U' = \coprod U'_i$ is a finite union of normal integral schemes and hence is normal.

As X' has an open covering by the schemes U' we conclude from Properties, Lemma 033J that X' is normal. On the other hand, each U' is a finite disjoint union of irreducible schemes, hence every quasi-compact open of X' has finitely many irreducible components (by a topological argument which we omit). Thus X' is a disjoint union of normal integral schemes by Properties, Lemma 0357. It is clear from the description of X' above that $Y \to X'$ is dominant and induces a bijection on irreducible components $V \to U'$ for every affine open $U \subset X$. The bijection of irreducible components for the morphism $Y \to X'$ follows from this by a topological argument (omitted). □